CUSTOMER SHOPPING BEHAVIOR ANALYSIS

A Data Analytics Project Report

Python (Pandas) | PostgreSQL | Power BI

PROJECT FOCUS
Understanding purchase patterns, product preferences, discount use, customer loyalty, shipping choices, and subscription behavior.

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Project materials reviewed: Python notebook, PostgreSQL SQL queries, Power BI dashboard file, and the supplied project report PDF

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Executive Summary

This project explores customer shopping behavior using a transactional dataset reported in the supplied project documentation as 3,900 purchase records and 18 original columns. The aim is to turn transaction and customer attributes into information that can support retail marketing, product decisions, discount planning, subscription growth, and customer retention.

The documented workflow uses Python and Pandas for data inspection, cleaning, and feature engineering; PostgreSQL for structured analysis; and Power BI for dashboard-based communication. The notebook shows category-level median imputation for missing review ratings, standardized column names, age-group creation, purchase-frequency mapping, and removal of the promo_code_used field after checking that it duplicates discount_applied in the data.

The SQL script contains ten business queries covering revenue by gender and age group, discounted high spenders, product ratings, shipping types, subscription behavior, discount rates, customer segments, category-level best sellers, and repeat buyers. The files supplied for review do not include readable result tables or validated numerical outputs for these queries. Therefore, this report documents the analysis design and recommendations without inventing findings.

1. Project Overview

Customer shopping datasets can reveal how spending and product preferences vary across customer groups and purchase contexts. This project establishes an analysis workflow that combines data preparation, SQL-based business questions, and a visual reporting layer.

1.1 Project Scope

- Examine customer demographics and purchase amounts.
- Review product categories, items purchased, ratings, and discount usage.
- Compare shopping behavior by shipping type and subscription status.
- Create customer segments using previous-purchase history.
- Present analysis through an interactive Power BI dashboard.

1.2 Core Business Question

How can the company use customer shopping data to identify patterns, improve engagement, and make better-informed marketing, product, and retention decisions?

2. Business Problem, Aim and Objectives

The project brief describes a retail business seeking a clearer understanding of purchasing patterns across customer demographics, product categories, discounts, reviews, seasons, and payment preferences. The intended outcome is a more evidence-led approach to sales, customer satisfaction, and long-term loyalty.

2.1 Objectives

1. Load and inspect the raw shopping dataset.
2. Identify and handle missing data where appropriate.
3. Standardize column names for consistent analysis.
4. Engineer age-group and purchase-frequency features.
5. Load the prepared dataset into PostgreSQL.

6. Answer ten defined business questions with SQL.
7. Communicate the analysis using a Power BI dashboard.
8. Develop practical recommendations linked to the analysis areas.

3. Dataset Description

Attribute	Description in supplied project material
Records	3,900 purchase records
Original columns	18 columns
Customer demographics	Age, gender, location, subscription status
Purchase details	Item purchased, category, purchase amount, season, size, colour
Shopping behavior	Discount applied, promo code used, previous purchases, purchase frequency, review rating, shipping type
Missing data	37 missing values in Review Rating, as documented in the report and notebook comments

The business brief mentions sales channels and payment preferences, but the supplied dataset summary does not list explicit fields for these attributes. Their availability should be checked before including channel- or payment-specific conclusions.

4. Tools and Technology Stack

Tool	Role	Activities evidenced in the supplied files
Python / Pandas	Preparation and transformation	Read the CSV, inspect structure and summary statistics, check missing values, standardize names, engineer fields
PostgreSQL	Structured analysis	Store the prepared table customer_shopping_data and run SQL business queries
Power BI	Visualization and reporting	Dashboard file supplied for interactive presentation of customer and purchase patterns
Jupyter Notebook	Reproducible workflow	Organize the data preparation and database-loading steps in notebook cells

5. Data Preparation and Feature Engineering

5.1 Loading and Initial Exploration

The notebook imports Pandas and loads a CSV named customer_shopping_behaviour.csv. It uses df.info() to review column types and non-null counts, df.describe() to inspect descriptive statistics, and df.isnull().sum() to count missing values.

5.2 Missing Review Ratings

The notebook comments identify 37 missing values in Review Rating. These are filled using the median review rating within each product category, implemented with a category group and a transform operation. This approach uses a category-specific median rather than one overall median.

5.3 Column Standardization

Column names are converted to lowercase and spaces are replaced with underscores. The purchase amount field is renamed from purchase_amount_(usd) to purchase_amount, providing a simpler name for SQL analysis.

5.4 Feature Engineering

- age_group: age is divided into four quantile-based groups and labelled Young Adult, Adult, Middle-aged, and Senior.
- purchase_frequency_days: purchase-frequency labels are mapped to approximate day values. The notebook maps weekly to 7, bi-weekly/fortnightly to 14, monthly to 30, quarterly/every 3 months to 90, and annually to 365.

5.5 Redundant Field Check

The notebook compares discount_applied and promo_code_used for exact equality. Its comments state that both fields carry the same information in this dataset, so promo_code_used is dropped. This decision should be revalidated if the source data changes, because a promotion code and a discount can represent different business concepts in other datasets.

6. Database Integration and SQL Analysis

The notebook uses SQLAlchemy with a PostgreSQL driver to connect to the local customer_behaviour database and write the prepared DataFrame to a table named customer_shopping_data. The SQL file then queries this table. Connection credentials are intentionally not reproduced in this report.

6.1 Business Questions Implemented

#	Analysis area	Question / measure
1	Revenue by gender	Sum purchase_amount by gender.
2	Discount users with high spend	Find customers who used a discount and spent at least the overall average purchase amount.
3	Highest-rated products	Rank the five items with the highest average review rating.
4	Shipping comparison	Compare average purchase amount for Standard and Express shipping.
5	Subscription comparison	Compare customer count, average spend, and total revenue for subscribers and non-subscribers.
6	Discount rate by product	Rank five products with the highest percentage of purchases where a discount was applied.
7	Customer segments	Classify records as New, Returning, or Loyal using previous_purchases, then count each segment.
8	Top products by category	Rank the three most purchased items within each category.
9	Repeat buyers and subscription	For customers with more than five previous purchases, count records by subscription status.
10	Revenue by age group	Sum purchase_amount by age_group and order by revenue.

6.2 Customer Segment Rules in SQL

Condition on previous_purchases	Segment label
Exactly 1	New
Between 2 and 10, inclusive	Returning
All other values	Loyal

Important implementation note: the SQL query counts rows in customer_shopping_data, which appears to contain purchase-level records. If each customer can have multiple rows, segment counts and subscription comparisons may represent records rather than unique customers. Validate the intended grain and consider COUNT(DISTINCT customer_id) where unique-customer counts are required.

7. Power BI Dashboard

A Power BI dashboard file was supplied as part of the project. The existing project report describes it as an interactive visual presentation of customer and purchase patterns. The dashboard is intended to help stakeholders explore the same analysis areas covered by the SQL questions.

Before final submission, confirm that each visual has a clear title, readable labels, consistent currency and percentage formats, appropriate filters, and measure definitions that match the SQL logic. Reconcile headline figures between Power BI and PostgreSQL so that the dashboard and report communicate consistent results.



8. Insight Interpretation Framework

The supplied SQL script defines the questions but does not provide the resulting values. The following interpretation guidance describes how to use those outputs once they have been run and validated; it is not a statement of observed results.

Analysis area	How to interpret the output
Gender and age-group revenue	Compare revenue alongside the number of customers or purchase records in each group; do not equate higher totals with higher individual spending.
Discount usage	Check whether discounted purchases coincide with higher purchase amounts and review discount-heavy products for margin impact.
Ratings and best sellers	Use average rating and purchase count together to distinguish well-reviewed items from frequently purchased items.
Shipping type	Compare average purchase amounts for Standard and Express; the comparison alone does not establish that shipping type causes spending differences.
Subscription and repeat purchasing	Compare average spend and revenue, then examine whether repeat purchase history is associated with subscription status.
Customer segments	Use segment sizes and behavior to tailor onboarding, repeat-purchase engagement, and loyalty initiatives.

9. Business Recommendations

These recommendations follow the analysis areas in the supplied project report and should be prioritized after the query results are validated.

- **Grow subscription adoption:** Communicate relevant subscriber benefits and measure adoption, repeat purchase behavior, and average spend before and after campaigns.
- **Strengthen loyalty:** Design engagement and rewards for repeat buyers, and track whether customers progress toward the defined Loyal segment.
- **Review discount policy:** Monitor products with high discount rates and evaluate profitability or margin alongside revenue where cost data is available.
- **Improve product positioning:** Use product ratings and purchase counts together when selecting items for campaigns and product discovery.
- **Target campaigns carefully:** Use validated age-group and customer-segment patterns to tailor messaging; avoid relying on revenue totals alone.
- **Evaluate shipping options:** Use Standard-versus-Express comparisons to understand observed purchase patterns, without claiming causation from a simple comparison.

10. Project Deliverables

- Python notebook for data inspection, cleaning, feature engineering, and PostgreSQL loading.
- PostgreSQL SQL script containing ten business queries.
- Power BI dashboard file for interactive analysis.
- Project report documenting the business context, workflow, analysis design, and recommendations.

11. Limitations and Future Improvements

- Run all SQL queries and insert verified result tables, figures, and concise evidence-based findings before treating the report as a results report.
- Confirm whether customer_id is unique per row or whether the dataset contains repeated transactions for customers; align all customer counts to the intended grain.
- Document the exact age boundaries produced by qcut, since quantile cut points depend on the dataset distribution.
- Validate the frequency mapping, especially synonyms such as Quarterly and Every 3 Months, and document how unrecognized labels or missing values are handled.
- Verify whether discount_applied and promo_code_used are identical across the full dataset before dropping one field in future refreshes.
- Check whether channel and payment preference fields exist before reporting on those business topics.
- Add profitability or margin measures if cost data becomes available, particularly when evaluating discount-heavy products.
- Record Power BI measure definitions, filter behavior, refresh date, and source-to-dashboard reconciliation checks.

Conclusion

The Customer Shopping Behavior Analysis project brings together a repeatable data workflow across Python, PostgreSQL, and Power BI. The notebook prepares the data and creates derived fields, the SQL script defines ten retail-focused questions, and the dashboard provides a visual reporting layer. Together, these assets form a

practical foundation for analyzing customer spending, product ratings, discount activity, shipping, subscriptions, repeat purchasing, and customer segments.

The next essential step is to run and validate the SQL queries, reconcile the results with the Power BI dashboard, and add quantified findings to the report. This will make the final document more actionable while ensuring that conclusions are supported by the project data.

Appendix A — SQL Business Questions

The supplied SQL file contains queries corresponding to the ten analysis topics listed in Section 6.1.

Implementation checks: verify the database connection and table schema; check whether review_rating is numeric; confirm that percentage calculations use decimal arithmetic where required; and ensure the output counts represent either customers or transactions as intended. The provided query for discount rate should be checked for integer division behavior in the configured PostgreSQL expression and cast to a numeric type if needed.

Appendix B — Source Files Reviewed

File	Use in this report
Customer_Shopping_Behavior_Analysis_Project_Report.pdf	Project context, dataset profile, documented methods, analysis areas, dashboard context, recommendations, and stated limitations.
Customer_Shopping_Behaviour_Analysis.ipynb	Notebook code for Pandas inspection, category-median imputation, column normalization, feature engineering, redundant-column check, and PostgreSQL loading.
customer_behaviour_sql_queries.sql	Ten SQL business questions and their query logic.
customer_behaviour_dashboard.pbix	Supplied Power BI dashboard artifact; its numerical visual outputs were not independently extracted into this report.

Evidence note: this report preserves the scope and terminology of the supplied project materials. It does not claim specific business results because readable query output values were not present in the reviewed files.